

Full Speaker Notes — Deep Learning-Based Planetary Crater Detection

Slide 1 — Title Slide

What This Slide Is Showing

This slide introduces the project title, course information, presenter name, and overall project topic.

What To Say

Hello everyone. My name is Scott Nguyen, and this presentation focuses on deep learning-based planetary crater detection using modern computer vision techniques.

The main goal of this project was to develop a complete crater detection pipeline capable of automatically identifying crater structures from planetary surface imagery.

Throughout this project, I explored multiple modern deep learning object detection architectures, including YOLOv8, Faster R-CNN, and RetinaNet.

I also developed supporting tools for:

- dataset verification,
- ground-truth visualization,
- model training,
- prediction generation,
- quantitative evaluation,
- and multi-model comparison.

Today I will discuss the motivation behind the project, the dataset and ground-truth pipeline, the deep learning models used, the training process, the experimental results, and the overall conclusions from the project.

Important Concepts

- Planetary crater detection
- Deep learning
- Object detection
- Computer vision
- Planetary imagery

Transition

First, I will discuss the motivation behind the project and explain why crater detection is an important computer vision problem.

Slide 2 — Outline

What This Slide Is Showing

This slide provides the overall structure of the presentation.

What To Say

This presentation is organized into several major sections.

First, I will discuss the introduction and project motivation.

Then I will introduce the crater dataset and explain the ground-truth annotation pipeline.

After that, I will discuss the deep learning architectures used throughout the project, including YOLOv8, Faster R-CNN, and RetinaNet.

Next, I will explain the training configuration and experimental methodology.

I will then present the quantitative and qualitative results, including model comparison metrics, training curves, precision-recall analysis, confusion matrices, and crater prediction examples.

Finally, I will discuss the strengths and limitations of the project, future work directions, and overall conclusions.

Transition

Now I will begin by discussing the motivation behind planetary crater detection.

Slide 3 — Introduction and Motivation

What This Slide Is Showing

This slide explains why planetary crater detection matters and why deep learning approaches are useful for this problem.

What To Say

Planetary crater detection is important for several scientific and engineering applications.

Crater distributions can provide information about:

- planetary surface age,
- geological evolution,
- impact history,
- terrain characterization,
- and surface morphology.

Crater analysis is commonly used in planetary science research for studying the Moon, Mars, and other planetary bodies.

Crater detection can also support future robotic exploration missions and autonomous planetary navigation systems.

Traditionally, crater identification is performed manually by human experts.

However, manual crater annotation becomes extremely time-consuming and difficult when dealing with very large planetary datasets.

This creates a strong motivation for automated crater detection using deep learning and computer vision methods.

The goal of this project was to build a complete deep learning-based crater detection pipeline and evaluate several modern object detection architectures.

Important Concepts

- Automated crater detection
- Planetary terrain analysis
- Deep learning object detection
- Computer vision

Transition

Next, I will discuss the specific objectives and deliverables of the project.

Slide 4 — Project Objectives

What This Slide Is Showing

This slide summarizes the major goals of the project.

What To Say

The project had several major objectives.

First, I wanted to build a complete crater detection workflow starting from dataset verification and ending with quantitative model evaluation.

Second, I needed to verify and visualize the crater ground-truth annotations to ensure that the bounding boxes were correct before training.

Third, I trained multiple modern deep learning object detection architectures including:

- YOLOv8,
- Faster R-CNN,
- and RetinaNet.

I then compared their crater detection performance using standard object detection metrics.

Another important goal was to generate qualitative crater prediction visualizations so the model behavior could be analyzed visually.

Finally, I explored limitations of the current system and potential future directions involving transformer-based architectures and foundation models.

Transition

Next, I will discuss the crater dataset used throughout the project.

Slide 5 — Dataset Overview

What This Slide Is Showing

This slide summarizes the crater dataset statistics and dataset structure.

What To Say

The project used a publicly available Martian and lunar crater detection dataset.

The dataset contains planetary surface images together with crater bounding-box annotations.

The annotations were stored using YOLO label format.

The dataset was divided into training, validation, and testing splits.

The training split contained:

- 98 images,
- and 681 crater annotations.

The validation split contained:

- 26 images,
- and 202 crater annotations.

The test split contained:

- 19 images,
- and 151 crater annotations.

Some images contained empty label files, meaning that no crater objects were annotated within those images.

Although the dataset is relatively small compared to large computer vision datasets, it was still sufficient for demonstrating crater localization learning behavior.

Important Concepts

- Training set
- Validation set
- Test set
- Bounding boxes
- YOLO labels

Transition

Next, I will explain how the crater ground-truth annotations were processed and visualized.

Slide 6 — Ground Truth Pipeline

What This Slide Is Showing

This slide explains how YOLO-format labels were converted into crater bounding-box visualizations.

What To Say

The crater annotations were stored using YOLO label format.

YOLO labels use normalized coordinates rather than absolute pixel coordinates.

That means the crater locations are represented relative to image size.

To visualize the crater annotations correctly, the normalized coordinates had to be converted into pixel-space coordinates.

The workflow shown here summarizes the full process.

First, the YOLO label files were loaded.

Next, the normalized crater coordinates were converted into pixel coordinates.

Then, crater bounding boxes were rendered directly onto the images.

Finally, the generated outputs were visually inspected to verify annotation quality.

Ground-truth verification is extremely important because incorrect annotations can significantly reduce object detection performance.

Important Concepts

- Ground truth
- Normalized coordinates
- Pixel coordinates
- Bounding boxes
- Annotation verification

Transition

Next, I will show an example of the crater ground-truth annotations.

Slide 7 — Ground Truth Example

What This Slide Is Showing

This slide shows a planetary image with crater bounding-box annotations.

What To Say

This figure shows an example of the crater ground-truth annotations used during model training and evaluation.

Each bounding box corresponds to a crater that was manually labeled within the dataset.

These labels provide the supervised learning targets used during training.

The purpose of these annotations is to teach the deep learning models where crater structures are located within the planetary imagery.

You can also observe several challenges within the image including:

- overlapping crater structures,
- varying crater sizes,
- complex terrain textures,
- and degraded crater boundaries.

These characteristics make crater detection a challenging computer vision problem.

Transition

Next, I will briefly discuss related crater detection research and prior methods.

Slide 8 — Related Work

What This Slide Is Showing

This slide summarizes previous crater detection methods and modern deep learning approaches.

What To Say

Earlier crater detection systems often relied on classical image processing techniques.

Examples include:

- edge detection,
- Hough transforms,
- template matching,
- and morphological filtering.

While these approaches can work under controlled conditions, they often struggle with:

- degraded crater rims,
- varying illumination,
- overlapping structures,
- and low-contrast terrain.

More recent crater detection approaches use deep learning and convolutional neural networks.

Modern object detection systems such as YOLO and Faster R-CNN have significantly improved crater localization accuracy and robustness.

Recent research also explores:

- transformer architectures,
- foundation models,
- and multimodal planetary vision systems.

This project focuses on comparing several modern deep learning object detection architectures for crater localization.

Transition

Next, I will explain the overall crater detection methodology used throughout the project.

Slide 9 — Overall Detection Pipeline

What This Slide Is Showing

This slide summarizes the full crater detection workflow.

What To Say

This slide shows the complete crater detection pipeline developed throughout the project.

The process begins with planetary surface images.

The crater ground-truth annotations are then verified and visualized.

Next, multiple deep learning object detection models are trained.

The trained models generate crater predictions on unseen test images.

Finally, the predictions are evaluated quantitatively using object detection metrics and compared across architectures.

This workflow allowed the entire crater detection system to be evaluated from dataset preparation through final model comparison.

Transition

Next, I will discuss the YOLOv8 architecture used throughout the project.

Slide 10 — YOLOv8

What This Slide Is Showing

This slide introduces the YOLOv8 object detection architecture.

What To Say

YOLOv8 is a modern one-stage object detection architecture.

A one-stage detector performs classification and localization simultaneously within a single network.

This makes YOLO architectures:

- fast,
- efficient,
- and highly effective for real-time object detection.

For this project, I used pretrained YOLOv8 nano weights.

Transfer learning was used so the model could leverage features learned from large-scale datasets.

The model was then fine-tuned specifically for crater detection.

YOLOv8 ultimately achieved the strongest overall crater detection performance in this project.

Important Concepts

- One-stage detector
- Transfer learning
- Bounding-box localization
- Real-time detection

Transition

Next, I will discuss Faster R-CNN and RetinaNet.

Slide 11 — Faster R-CNN and RetinaNet

What This Slide Is Showing

This slide introduces the Faster R-CNN and RetinaNet architectures.

What To Say

Faster R-CNN is a two-stage object detector.

The first stage generates region proposals.

The second stage performs classification and bounding-box refinement.

This architecture often provides strong localization accuracy but is generally more computationally expensive.

RetinaNet is another one-stage detector that uses focal loss to improve performance under class imbalance conditions.

Both models used pretrained ResNet50 Feature Pyramid Network backbones.

The crater annotations had to be converted from YOLO format into PyTorch-compatible bounding-box tensors before training.

Important Concepts

- Two-stage detection
- Region Proposal Network
- Focal loss
- ResNet50-FPN backbone

Transition

Next, I will discuss the training configuration used throughout the experiments.

Slide 12 — Training Configuration

What This Slide Is Showing

This slide summarizes the training settings used for the deep learning models.

What To Say

All models were trained using transfer learning.

YOLOv8 was trained using the Ultralytics framework.

Faster R-CNN and RetinaNet were trained using PyTorch torchvision.

YOLOv8 was trained for 50 epochs.

Faster R-CNN and RetinaNet were trained for 3 epochs each.

Different batch sizes were used depending on architecture complexity.

The evaluation metrics included:

- precision,
- recall,
- mAP at 0.5,
- and mAP at 0.5 to 0.95.

These metrics are standard object detection evaluation metrics.

mAP stands for mean Average Precision and measures how accurately the model localizes objects.

Transition

Next, I will present the quantitative crater detection results.

Slide 13 — Quantitative Results

What This Slide Is Showing

This slide presents the final crater detection performance metrics.

What To Say

This table summarizes the final quantitative crater detection results.

YOLOv8 achieved the strongest overall performance with an mAP at 0.5 of approximately 0.70.

Faster R-CNN achieved very similar crater localization performance.

RetinaNet performed significantly worse under the current experimental configuration.

Several factors likely contributed to the RetinaNet performance reduction including:

- limited dataset size,
- reduced training duration,
- and optimization difficulty.

The results demonstrate that YOLOv8 and Faster R-CNN were both capable of learning meaningful crater localization behavior.

Important Concepts

- mAP
- Precision
- Recall
- Localization accuracy

Transition

Next, I will discuss the YOLOv8 training curves.

Slide 14 — Training Curves

What This Slide Is Showing

This slide shows the YOLOv8 training convergence behavior.

What To Say

This figure shows the YOLOv8 training curves generated during training.

The curves show how the training and validation metrics evolved throughout the training process.

You can observe relatively stable convergence behavior.

As training progressed, the validation metrics improved steadily.

This indicates that the model successfully learned crater localization features from the planetary imagery dataset.

Stable convergence behavior is important because unstable training can indicate:

- overfitting,
- optimization problems,
- or insufficient training data.

Transition

Next, I will discuss the precision-recall analysis.

Slide 15 — Precision-Recall Curve

What This Slide Is Showing

This slide presents the YOLOv8 precision-recall curve.

What To Say

This figure shows the precision-recall curve generated during evaluation.

Precision measures how many predicted craters were actually correct.

Recall measures how many true crater objects were successfully detected.

The curve summarizes the tradeoff between false positives and missed detections.

The generated curve demonstrates relatively strong precision-recall behavior across multiple operating thresholds.

This indicates that the detector maintained relatively accurate crater predictions while still detecting many crater structures.

Important Concepts

- Precision
- Recall
- False positives
- Missed detections

Transition

Next, I will discuss the F1 curve and confusion matrix.

Slide 16 — F1 Curve and Confusion Matrix

What This Slide Is Showing

This slide shows the F1-score curve and normalized confusion matrix.

What To Say

The F1 score combines precision and recall into a single metric.

A higher F1 score generally indicates better balance between accurate detections and successful crater coverage.

The confusion matrix provides additional information about classification performance.

It shows how many crater instances were classified correctly versus incorrectly.

The normalized confusion matrix indicates that the YOLOv8 detector achieved relatively strong crater classification capability with low false-positive behavior.

Transition

Next, I will show qualitative crater prediction examples.

Slide 17 — Qualitative Detection Results

What This Slide Is Showing

This slide compares crater ground truth against YOLOv8 predictions.

What To Say

This slide shows a side-by-side comparison between crater ground-truth annotations and YOLOv8 predictions.

The left image shows the manually labeled crater annotations.

The right image shows crater detections generated by the trained YOLOv8 model.

The detector successfully localized many crater structures under varying terrain conditions.

However, several challenges remain including:

- overlapping craters,
- small crater localization,
- degraded crater rims,
- and low-contrast terrain regions.

Even under these conditions, the detector demonstrated relatively strong localization capability.

Transition

Next, I will summarize the overall model comparison.

Slide 18 — Detector Output Comparison

What This Slide Is Showing

This slide compares the same test image across four outputs: the ground truth, YOLOv8 prediction, Faster R-CNN prediction, and RetinaNet prediction.

What To Say

This slide is meant to visually compare how each detector behaved on the same crater image.

The top-left image shows the ground truth. These are the labeled crater boxes from the dataset. This represents what the model is supposed to learn.

The top-right image shows the YOLOv8 prediction. YOLOv8 produced the cleanest and strongest detection behavior overall. It found crater regions with relatively good localization and fewer unnecessary boxes compared with Faster R-CNN.

The bottom-left image shows the Faster R-CNN prediction. Faster R-CNN did detect many crater regions, which agrees with the quantitative results showing that Faster R-CNN had an mAP@0.5 close to YOLOv8. However, the prediction appears less clean because it produced many detections. This suggests that Faster R-CNN was able to identify crater-like structures, but its confidence filtering and localization behavior were not as clean as YOLOv8 on this example.

The bottom-right image shows the RetinaNet prediction. At a confidence threshold of 0.3, RetinaNet produced no confident crater detections. This is not a plotting mistake. It is consistent with the quantitative results, where RetinaNet had a near-zero mAP score.

The main point of this slide is that the qualitative output matches the numerical results. YOLOv8 performed best, Faster R-CNN was competitive but less clean, and RetinaNet struggled to learn useful crater detections under the current limited-data training setup.

Important Concepts

- Ground truth: the correct labeled crater boxes.
- Prediction: the model-generated crater boxes.
- Confidence threshold: the minimum confidence required before drawing a detection box.
- Qualitative comparison: visual comparison of model outputs.
- Quantitative comparison: metric-based comparison such as mAP.

Key Point To Emphasize

At the same confidence threshold, YOLOv8 and Faster R-CNN produced crater detections, while RetinaNet produced no confident detections. This supports the final model comparison table and plot.

Transition

Next, I will summarize the final quantitative comparison across all models.

Slide 19 — Final Model Comparison

What This Slide Is Showing

This slide summarizes the final multi-model comparison.

What To Say

This figure summarizes the final crater detection performance comparison across all evaluated architectures.

YOLOv8 achieved the strongest overall crater localization performance.

Faster R-CNN achieved competitive performance but required a more computationally complex two-stage architecture.

RetinaNet underperformed under the current limited-data training conditions.

The results demonstrate that architecture selection plays an important role in crater detection performance.

The results also demonstrate that deep learning object detection frameworks are capable of learning meaningful crater localization behavior even with relatively small datasets.

Transition

Next, I will discuss the strengths, limitations, and future directions of the project.

Slide 20 — Discussion

What This Slide Is Showing

This slide discusses interpretation of the results.

What To Say

YOLOv8 likely achieved the strongest performance because of:

- efficient one-stage detection design,
- strong optimization behavior,
- and effective transfer learning.

Faster R-CNN also demonstrated strong localization capability because of its region proposal refinement stage.

RetinaNet likely struggled because focal-loss optimization can become difficult under limited-data conditions.

The crater detection problem itself is also very challenging due to:

- varying crater sizes,
- degraded crater rims,
- overlapping structures,
- illumination variability,
- and terrain ambiguity.

The project also demonstrated the importance of:

- annotation quality,
- dataset preparation,
- model selection,
- and training configuration.

Transition

Next, I will discuss future work directions.

Slide 21 — Future Work

What This Slide Is Showing

This slide discusses possible future extensions.

What To Say

Several future research directions could improve crater detection performance.

One important improvement would involve using larger planetary imagery datasets.

Additional directions include:

- transformer-based detectors,
- crater segmentation methods,
- semi-supervised learning,
- and foundation-model-based crater detection systems.

Foundation models are especially interesting because they can leverage large-scale pretrained vision representations.

Future work could also explore:

- multi-class geological feature detection,

- domain adaptation,
- and higher-resolution imagery.

Transition

Finally, I will conclude the presentation.

Slide 22 — Conclusion

What This Slide Is Showing

This slide summarizes the overall project contributions.

What To Say

In conclusion, this project developed a complete deep learning-based planetary crater detection pipeline.

The system incorporated:

- dataset verification,
- ground-truth visualization,
- model training,
- crater prediction generation,
- and quantitative evaluation.

Multiple modern object detection architectures were evaluated.

YOLOv8 achieved the strongest overall crater detection performance under the current experimental configuration.

The results demonstrated that deep learning object detection methods are highly promising for automated planetary terrain analysis.

Overall, the project demonstrated the importance of:

- dataset quality,
- annotation consistency,
- model architecture selection,
- and training configuration.

Thank you.

Likely Questions

Why did RetinaNet perform poorly?

RetinaNet likely underperformed because the crater dataset was relatively small and the model was only trained for a limited number of epochs. RetinaNet relies heavily on focal-loss optimization, which can become difficult under limited-data conditions. In contrast, YOLOv8 converged more effectively under the current training setup.

Why use YOLO labels?

YOLO label format is lightweight, widely supported, and commonly used for object detection tasks. The dataset already provided crater annotations in YOLO format, which made it efficient to train YOLOv8 directly and convert the annotations into PyTorch-compatible bounding boxes for Faster R-CNN and RetinaNet.

Why was YOLOv8 the strongest model?

YOLOv8 benefited from a highly optimized one-stage detection architecture, efficient feature extraction, and strong transfer-learning support from pretrained weights. The Ultralytics training framework also provides stable optimization and augmentation pipelines, which likely improved crater localization performance.

Could segmentation improve results?

Yes. Segmentation models could potentially improve crater boundary localization because they predict pixel-level crater regions instead of only rectangular bounding boxes. This could help with overlapping craters, degraded rims, and irregular crater shapes.

What would happen with a larger dataset?

A larger dataset would likely improve generalization performance and reduce overfitting. More crater examples would expose the models to a wider variety of crater sizes, terrain textures, and illumination conditions, which would likely improve robustness.

Could foundation models help?

Yes. Foundation models and transformer-based vision systems are promising because they are pretrained on very large image datasets and can transfer learned visual features to new tasks. These models may improve crater detection performance, especially under limited-data conditions.

Why is crater detection difficult?

Crater detection is challenging because crater structures vary significantly in size, shape, illumination, and degradation level. Some craters overlap with each other, while others blend into terrain textures or have partially eroded rims.

Why use transfer learning?

Transfer learning allows the models to start from pretrained visual features rather than learning everything from scratch. This is especially important when working with smaller datasets because it improves convergence speed and overall performance.

What does mAP mean?

mAP stands for mean Average Precision. It is a standard object detection metric that measures how accurately a model both localizes and classifies objects. Higher mAP values indicate stronger detection performance.

What is the difference between precision and recall?

Precision measures how many predicted craters were actually correct. Recall measures how many real craters were successfully detected. A strong detector should ideally achieve both high precision and high recall.

Why compare multiple architectures?

Different object detection architectures have different strengths and weaknesses. Comparing multiple models helps determine which architecture is most effective for crater localization under the current dataset and training conditions.